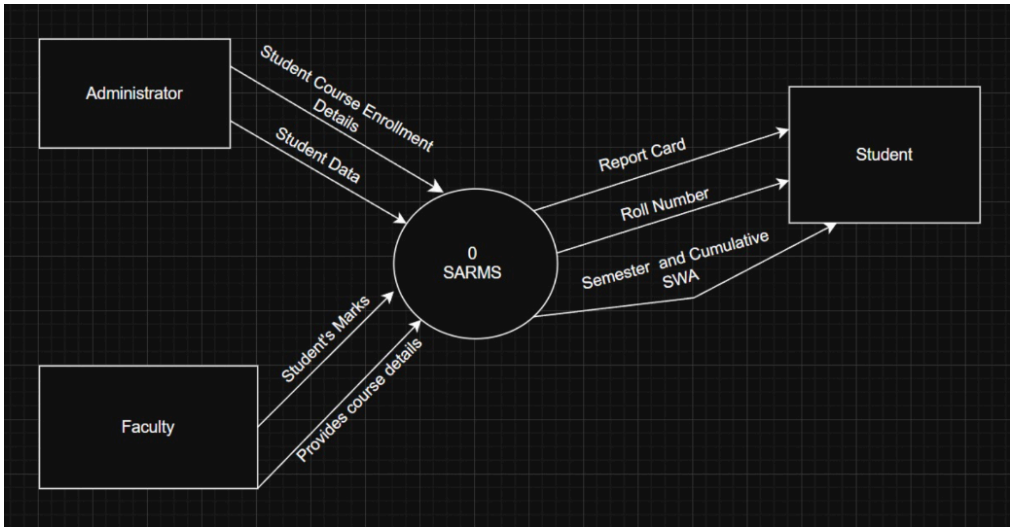
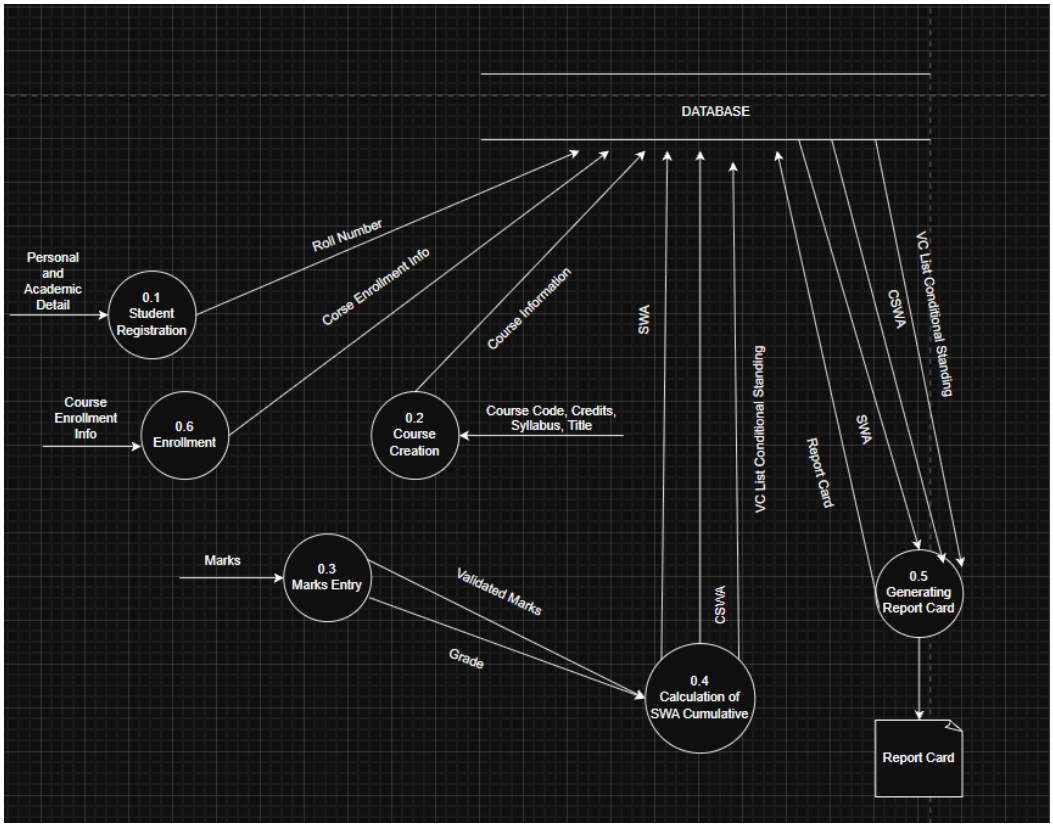
GROUP 9
DFD Model and Data Dictionary

DFD MODEL

Level 0

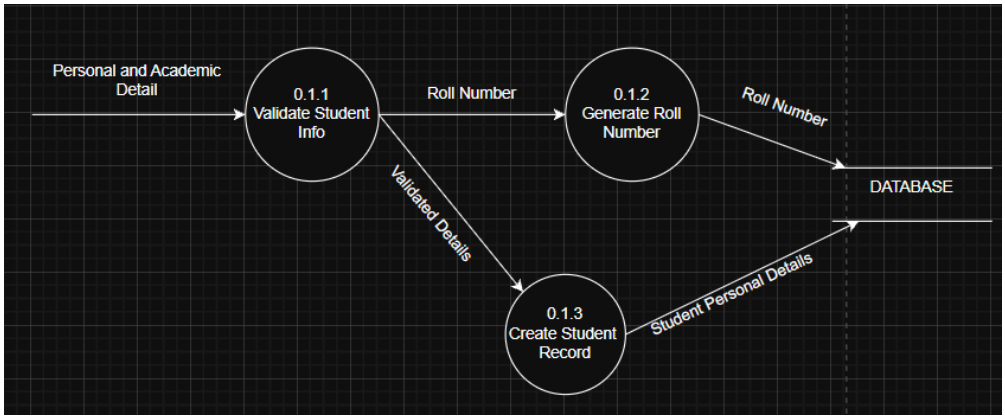


Level 1



Level 2

0.1



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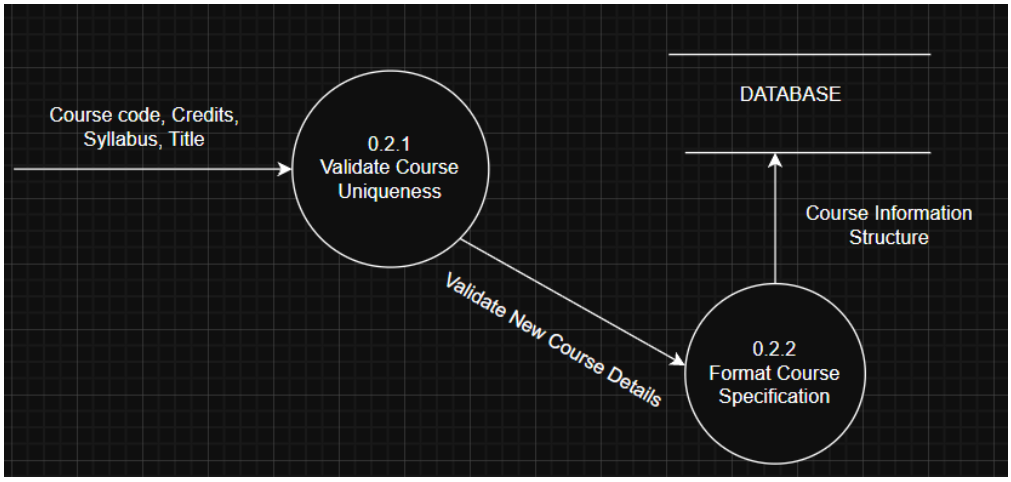

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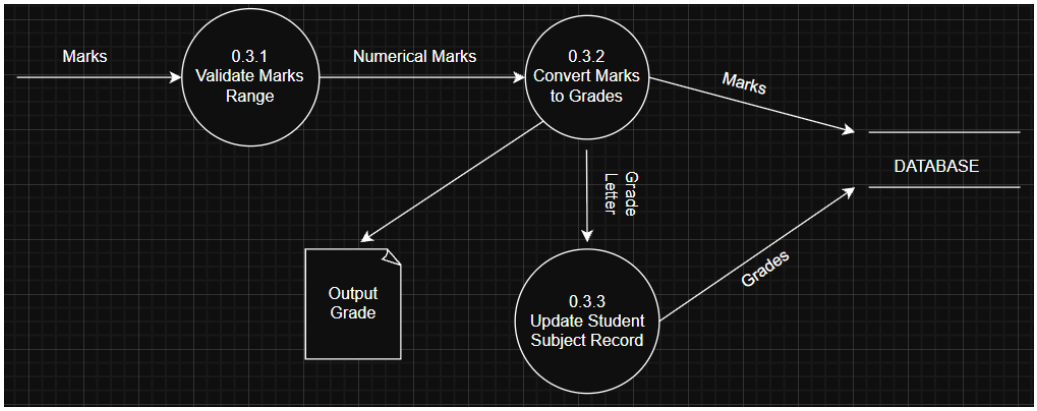

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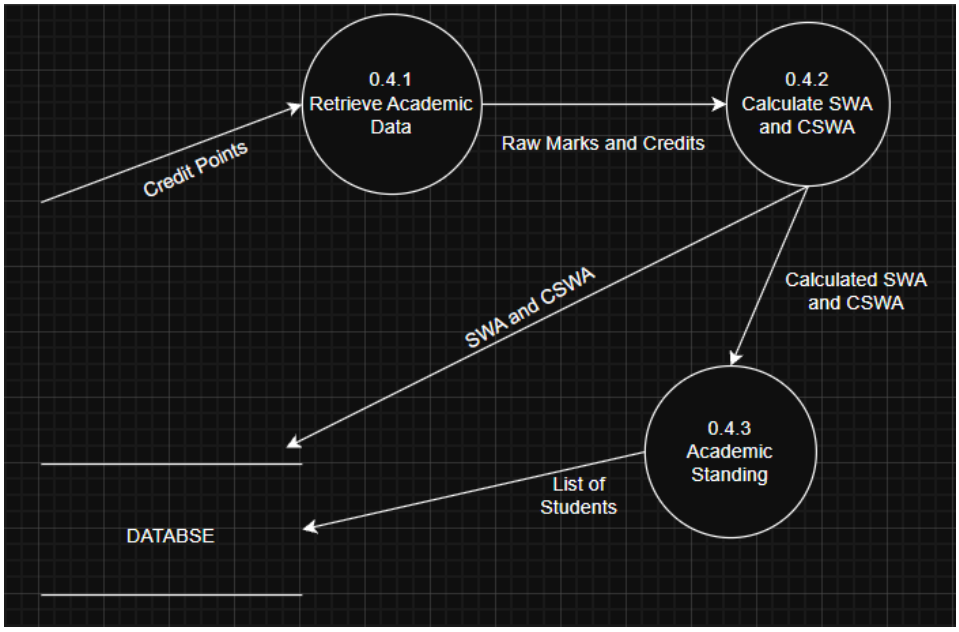
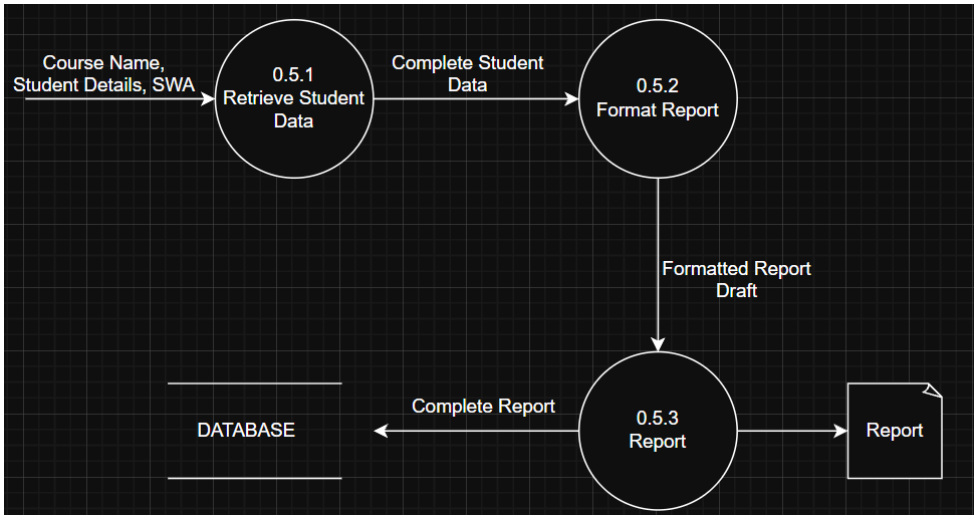
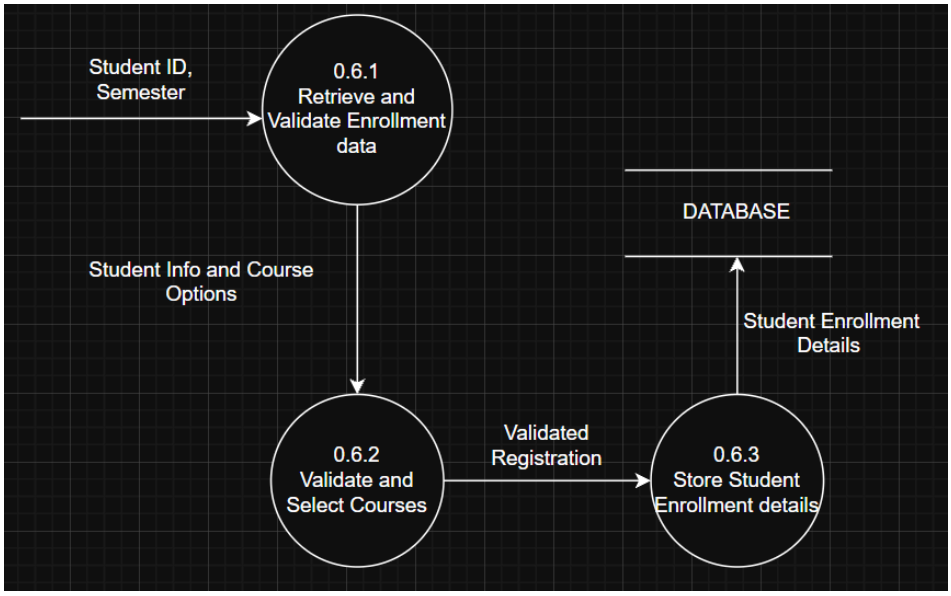

0.4.1
Retrieve Academic Data
0.4.2
Calculate SWA and CSWA
Credit Points
Raw Marks and Credits
SWA and CSWA
Calculated SWA and CSWA
0.4.3
Academic Standing
List of Students
DATABSE

0.5


Course Name, Student Details, SWA
0.5.1
Retrieve Student Data
Complete Student Data
0.5.2
Format Report
Formatted Report Draft
DATABASE
Complete Report
0.5.3
Report
Report

0.6


Student ID, Semester
0.6.1 Retrieve and Validate Enrollment data
DATABASE
Student Info and Course Options
Student Enrollment Details
0.6.2 Validate and Select Courses
Validated Registration
0.6.3 Store Student Enrollment details

Data Dictionary

Level 0 Data Dictionary

STUDENT DATA

STUDENT DATA = NAME + ADDRESS + CONTACT NO+ SEMESTER NO

STUDENT COURSE ENROLLMENT DETAILS

STUDENT COURSE ENROLLMENT DETAILS = ROLL NO + SEMESTER NO+ COURSE CODE

STUDENT MARKS

STUDENT MARKS = ROLL NO +COURSE CODE + MARKS

COURSE DETAILS

COURSE DETAILS = COURSE CODE+ COURSE TITLE + CREDITS + SYLLABUS

ROLL NUMBER

ROLL NUMBER = ROLL NO

SEMESTER AND CUMULATIVE SWA

SEMESTER AND CUMULATIVE SWA = SWA+ CGPA

REPORT CARD

REPORT CARD = STUDENT DETAILS +STUDENT SUBJECT RECORD+ SWA+ CGPA + ACADEMIC STATUS

Level 1 Data Dictionary

STUDENT REGISTRATION

STUDENT REGISTRATION = PERSONAL AND ACADEMIC DETAIL+ ROLL NUMBER

COURSE CREATION

COURSE CREATION = COURSE INFORMATION

MARKS ENTRY

MARKS ENTRY = VALIDATED MARKS + GRADE

CALCULATION OF SWA AND CUMULATIVE

SWA = Σ(MARKS × CREDITS) / Σ(CREDITS)

CGPA = Σ(SWA) / TOTAL SEMESTERS

GENERATING REPORT CARD

GENERATING REPORT CARD = REPORT DATA

ENROLLMENT

ENROLLMENT = COURSE ENROLLMENT INFO

Level 2 Data Dictionary

0.1 Student Registration Decomposition

VALIDATE STUDENT INFO

VALIDATE STUDENT INFO = PERSONAL AND ACADEMIC DETAIL

GENERATE ROLL NUMBER

GENERATE ROLL NUMBER = ROLL NO

CREATE STUDENT RECORD

CREATE STUDENT RECORD = VALIDATED DETAILS
+ ROLL NO

0.2 Course Creation Decomposition

VALIDATE COURSE UNIQUENESS

VALIDATE COURSE UNIQUENESS = COURSE CODE + CREDITS+ SYLLABUS+COURSE TITLE

FORMAT COURSE SPECIFICATION

FORMAT COURSE SPECIFICATION = COURSE INFORMATION

0.3 Marks Entry Decomposition

VALIDATE MARKS RANGE

VALIDATE MARKS RANGE = MARKS

NUMERICAL MARKS

NUMERICAL MARKS = ROLL NO+ COURSE CODE+ MARKS

CONVERT MARKS TO GRADES

CONVERT MARKS TO GRADES = LETTER GRADE

UPDATE STUDENT SUBJECT RECORD

UPDATE STUDENT SUBJECT RECORD = ROLL NO+ COURSE CODE+ MARKS+ LETTER GRADE

0.4 SWA and CSWA Calculation Decomposition

RETRIEVE ACADEMIC DATA

RETRIEVE ACADEMIC DATA = CREDIT POINTS+ RAW MARKS

SWA = Σ(MARKS × CREDITS) / Σ(CREDITS)

CSWA = Σ(SWA) / TOTAL SEMESTERS

ACADEMIC STANDING

ACADEMIC STANDING = VC LIST+ CONDITIONAL+ NORMAL

0.5 Report Generation Decomposition

RETRIEVE STUDENT DATA

RETRIEVE STUDENT DATA = COURSE NAME+ STUDENT DETAILS+ SWA

FORMAT REPORT

FORMAT REPORT = COMPLETE STUDENT DATA

FINAL REPORT

FINAL REPORT = FORMATTED REPORT

0.6 Enrollment Decomposition

RETRIEVE AND VALIDATE ENROLLMENT DATA

RETRIEVE AND VALIDATE ENROLLMENT DATA = STUDENT ID+ SEMESTER

VALIDATE AND SELECT COURSES

VALIDATE AND SELECT COURSES = STUDENT INFO+ COURSE OPTIONS

STORE STUDENT ENROLLMENT DETAILS

STORE STUDENT ENROLLMENT DETAILS = VALIDATED REGISTRATION

Primitive Data Elements

Data Element	Type
ROLL NO	Integer
NAME	String
ADDRESS	String
CONTACT NO	Numeric
SEMESTER NO	Integer
COURSE CODE	String
COURSE TITLE	String
CREDITS	Integer
SYLLABUS	String
MARKS	Integer (0–100)
LETTER GRADE	Character
SWA	Float
CGPA	Float
ACADEMIC STATUS	String

Notation Used

"=" : Defined as

"+" : Composition

"{ }": Zero or more occurrences

"[a,b]": Either a or b